Exploring Enceladus: The Science Case for Future Radar Sounder Measurements

Hauke Hussmann[1], William Byrne[1,2], Ana-Catalina Plesa[1], Tina Rückriemen-Bez[1,3], Andreas Benedikter[4]

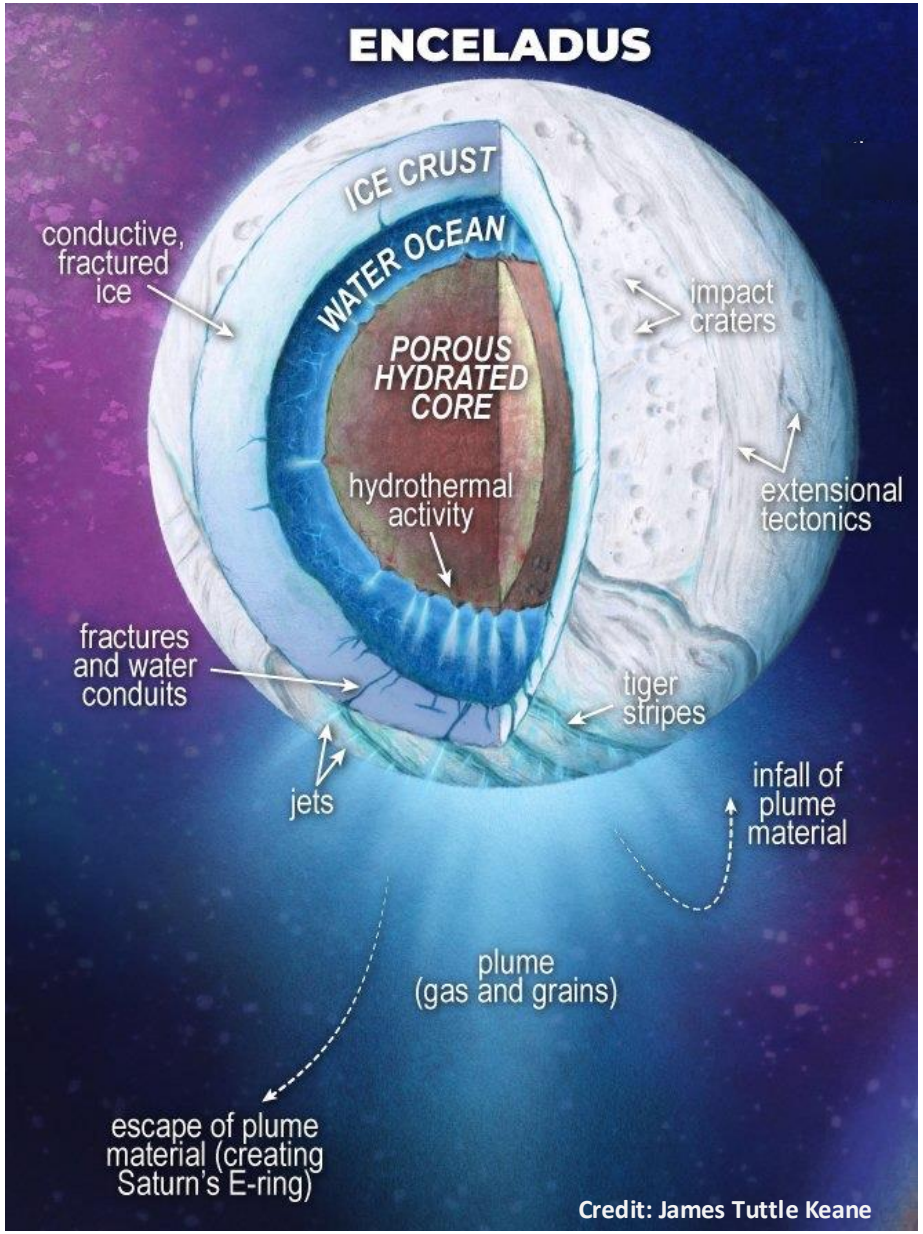
[1]German Aerospace Center (DLR), Institute of Planetary Research
[2]Freie Universität Berlin, Institute of Geological Sciences
[3]University of Münster, Institute for Planetology
[4]German Aerospace Center (DLR), Microwaves and Radar Institute

DLR

Enceladus

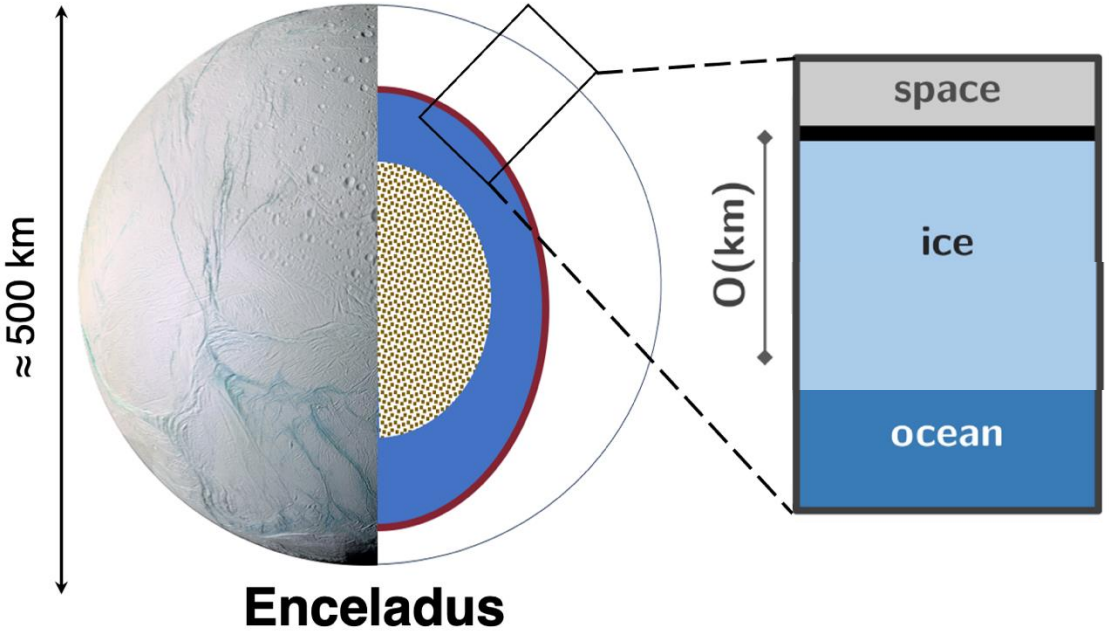
- Overarching Question:
  - Where do we find conditions conducive for life to evolve?
  - Where do we find liquid water at stable conditions?

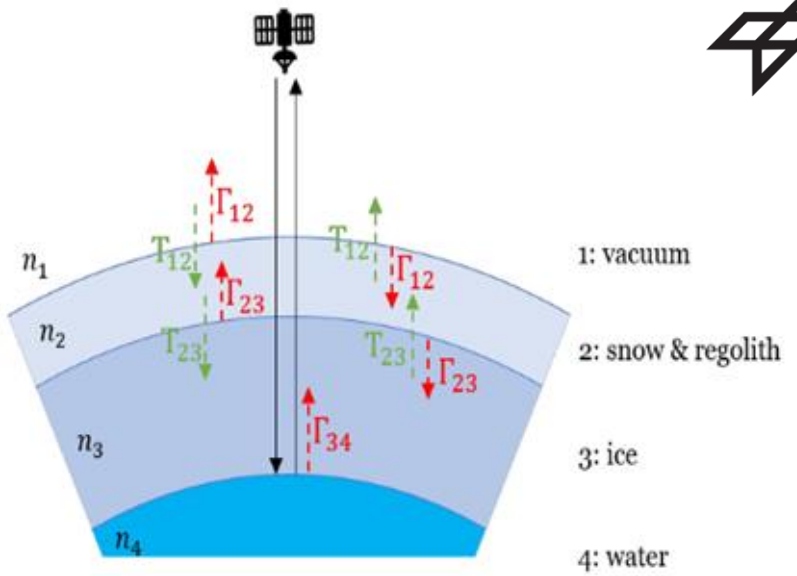
- Saturn's moon Enceladus:
  - liquid water (global oceans and local reservoirs)
  - the 'right' chemistry (rock-water interaction)
  - energy (tidal friction)
  - stable conditions (liquid water at constant temperatures)
  - dynamical processes/cycles for material exchange (active satellites)



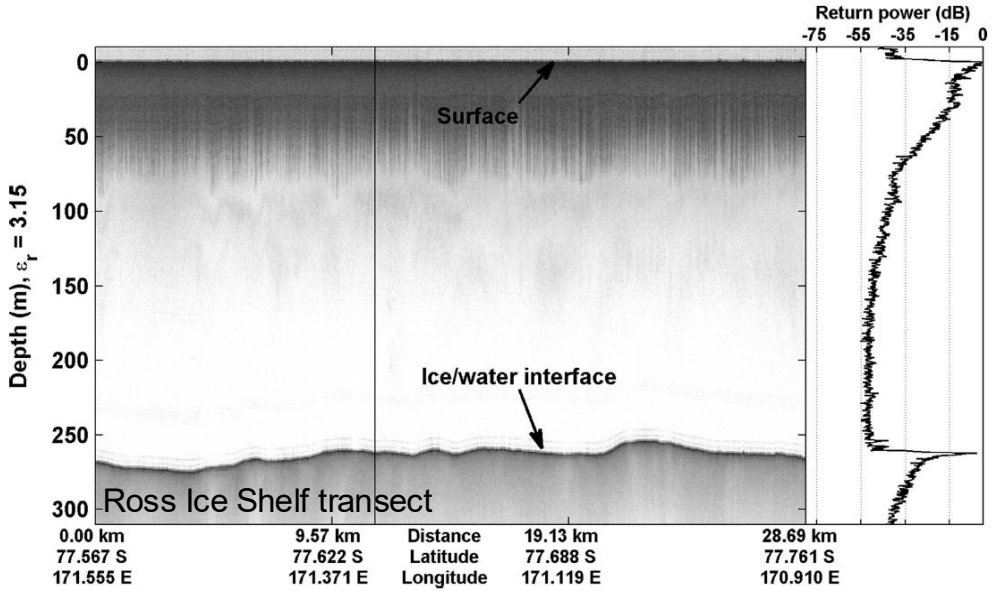
Enceladus ice shell

- How thick is the ice shell?
- What is the thermal, physical, and chemical state of the ice shell?
- How do ice shell properties and parameters affect the 2-way radar attenuation?





Detection of ice-water interface on Earth



Lewis et al. 2015

Radar Sounding: 2-way Radar Attenuation

- The more material the signal goes through, the more “attenuated” the signal becomes
- Temperature and impurities are driving attenuation

Typical assumption:

- Estimated attenuation limit: 100 dB

[Kalousová et al., 2017; Souček et al., 2023]

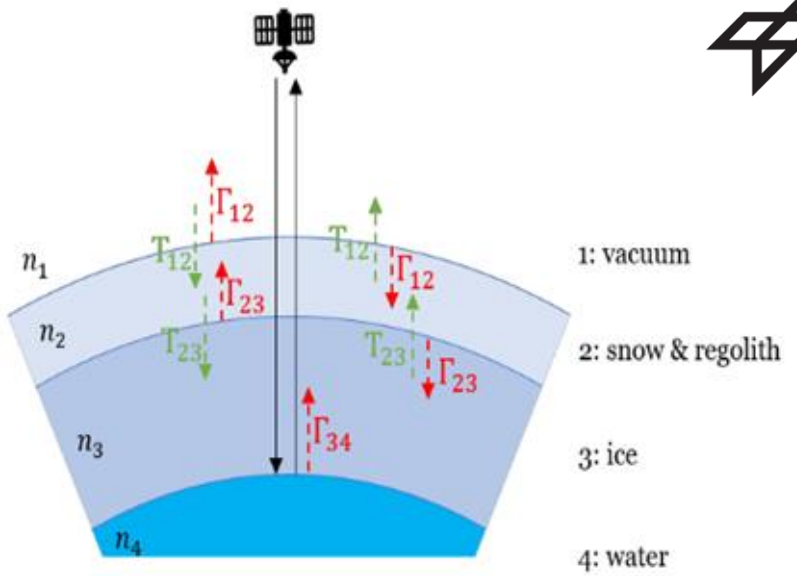
- Ice shell most-likely conductive

[Montagnat & Duval, 2000]

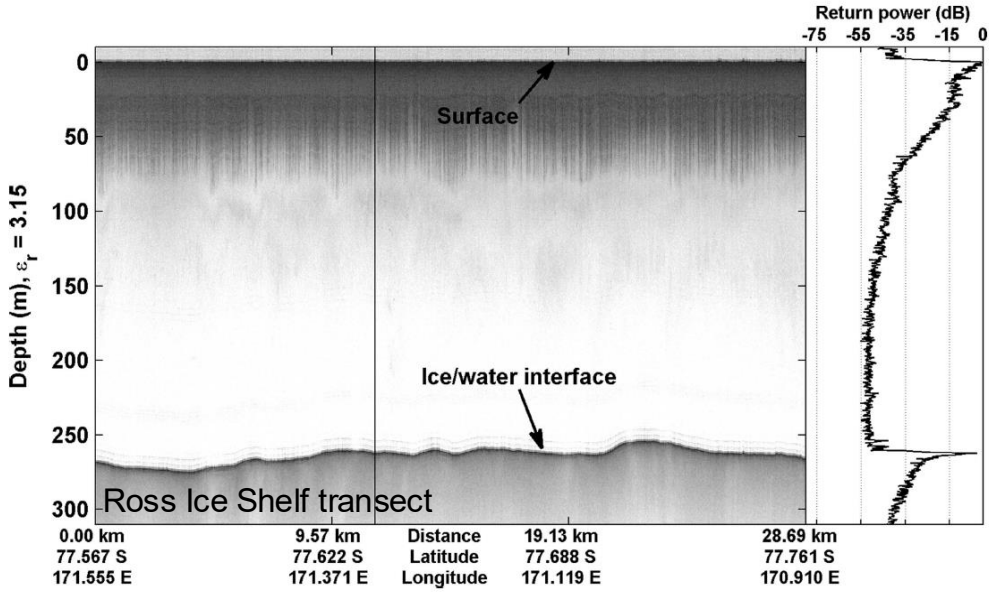
See poster # X4.127 by DeMers et al.

- Material fall-back from the Enceladus’ plume eruptions leads to porous layer thicknesses of up to 700 m

[Martin et al., 2023]



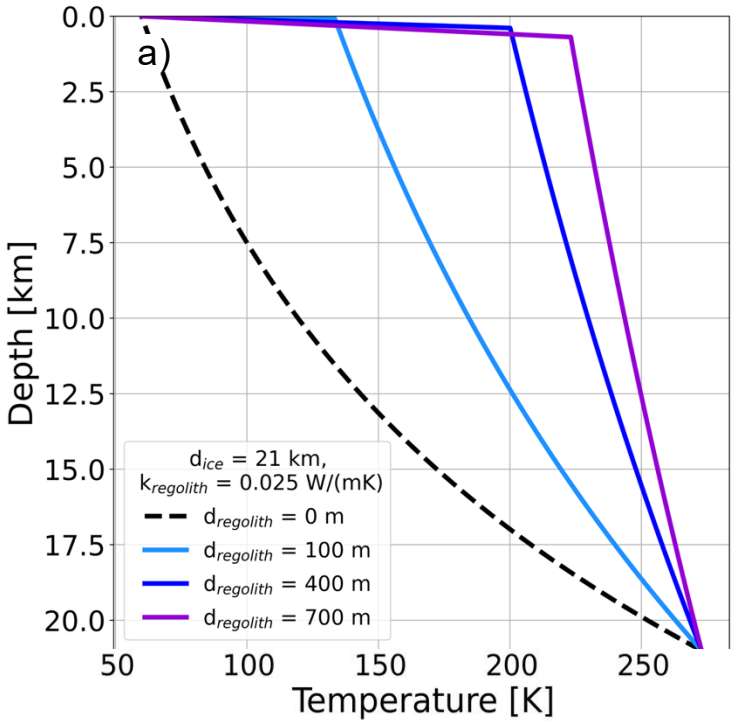
Detection of ice-water interface on Earth



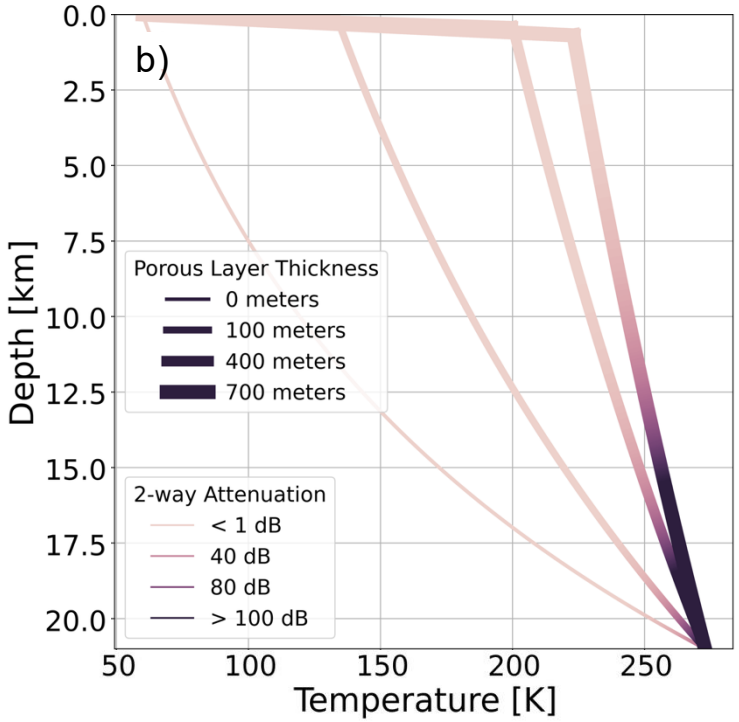
Lewis et al. 2015

Effects of Porous Layer on 2-way Attenuation

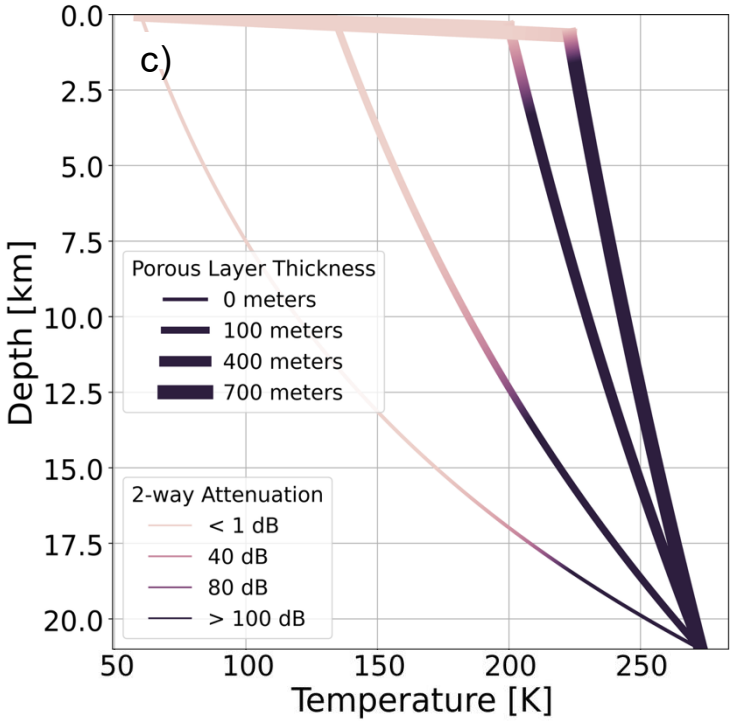
- A porous layer keeps the subsurface warm
- The thicker the porous layer the higher the subsurface temperature is
- The higher the suubsurface temeprature the higher the 2-way attenuation becomes



Low-loss case (chloride concentration: 0 μM)



High-loss case (chloride concentration: 300 μM)



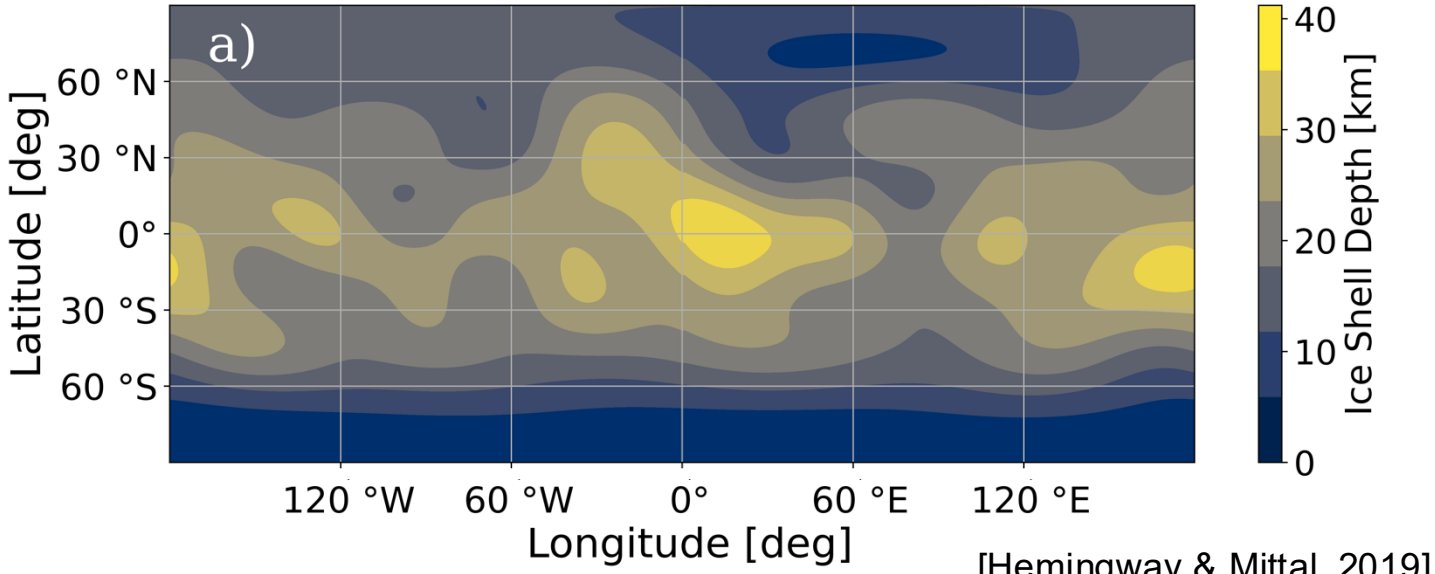
[Byrne et al., JGR:Planets, 2024]

Enceladus: Ice Shell Thickness Variations

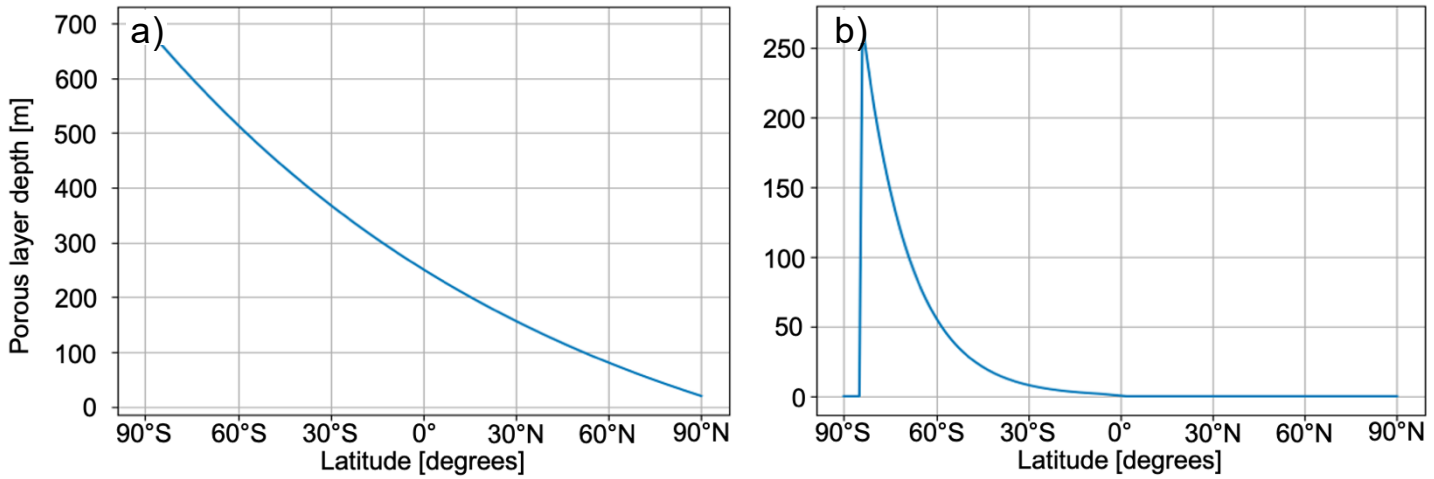
- The ice shell thickness: obtained from shape*, gravity and libration data

- Ice shell thickness:
  - Average value: 21-23 km
  - South pole:
    values as low as 5 km
  - Equatorial regions:
    values as high as 35 km

- Porous layer:
  - Thickness up to 700 m
  - Distribution poorly constrained

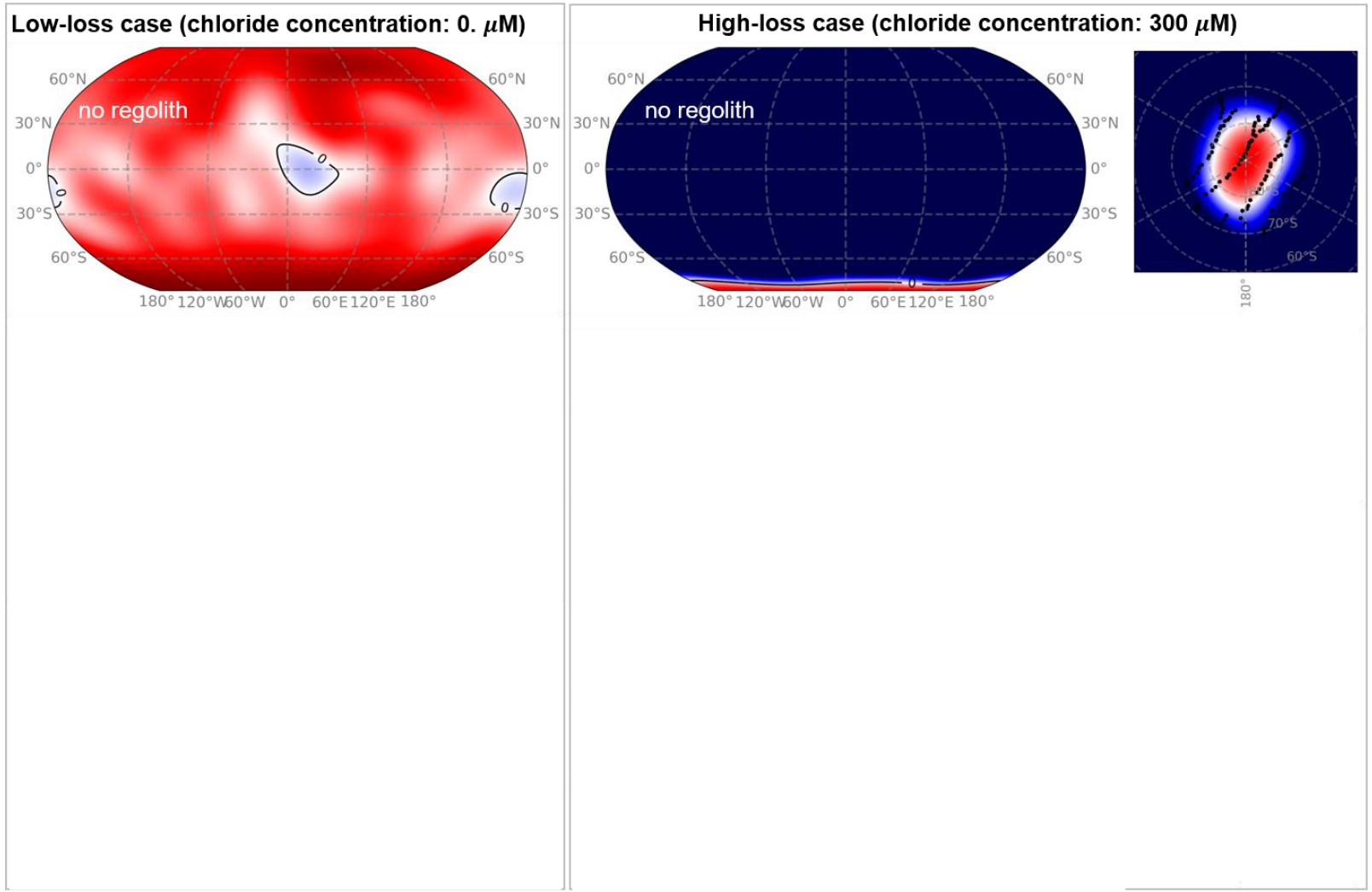


[Hemingway & Mittal, 2019]



Effects of Porous Layer Distribution on 2-way Attenuation

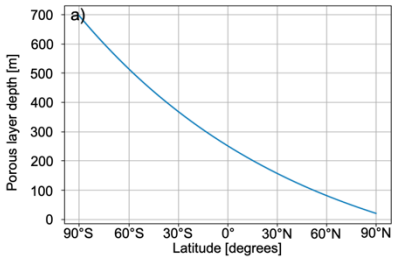
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- Most extreme case: detection of the ice-ocean interface expected only in polar areas

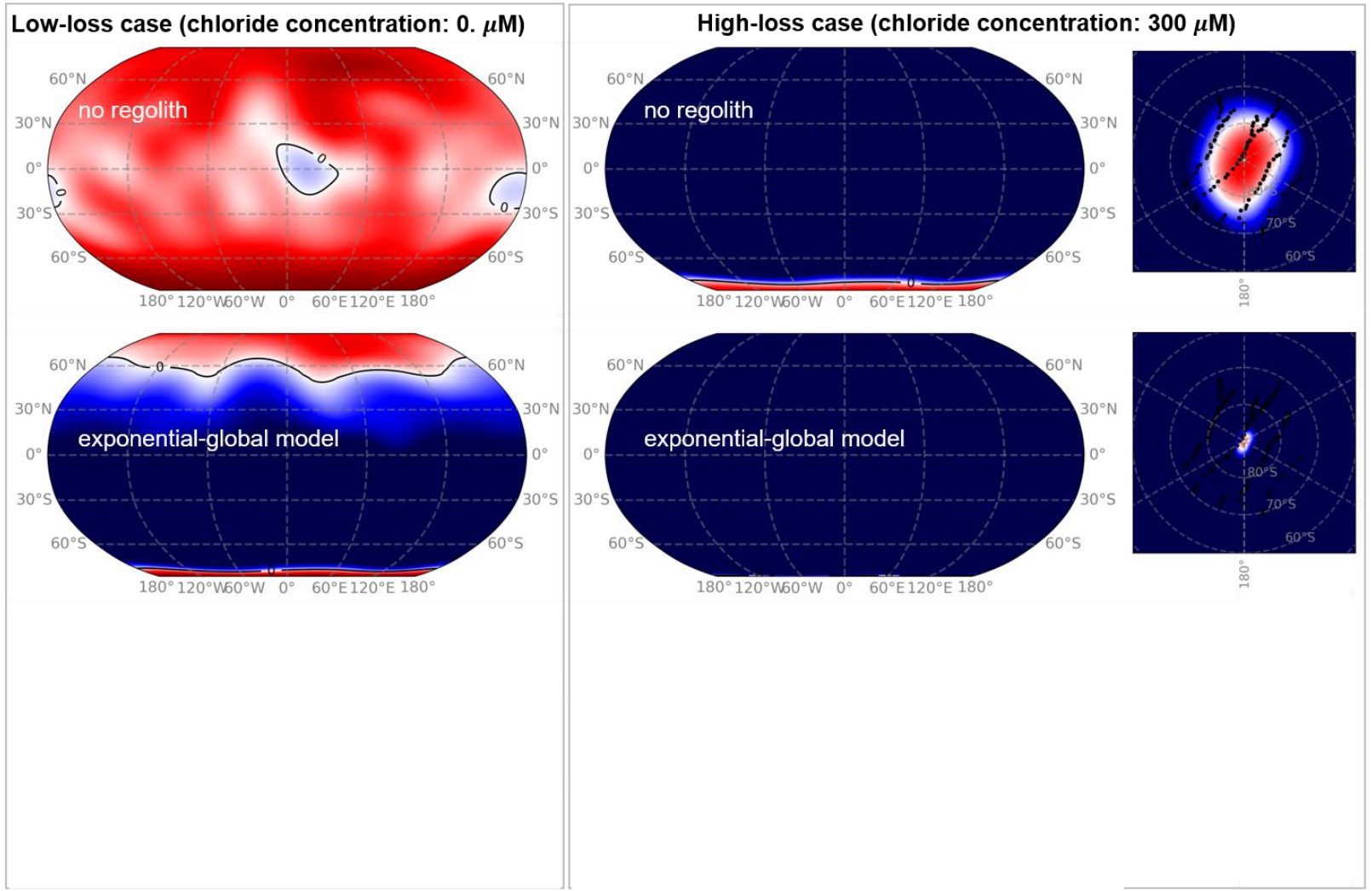


*assuming avg transmit power of 1.25 W and 100 km altitude orbit

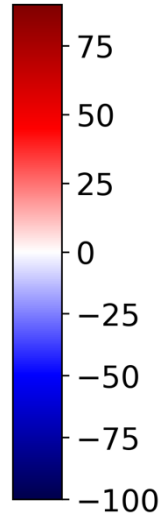
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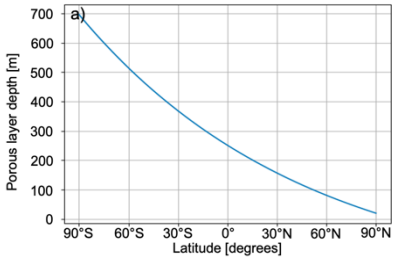
SNR [dB]
ice-ocean interface

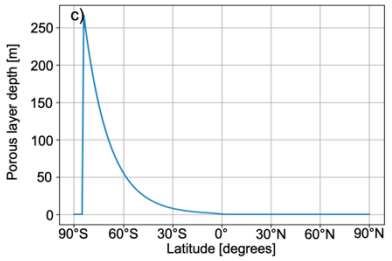


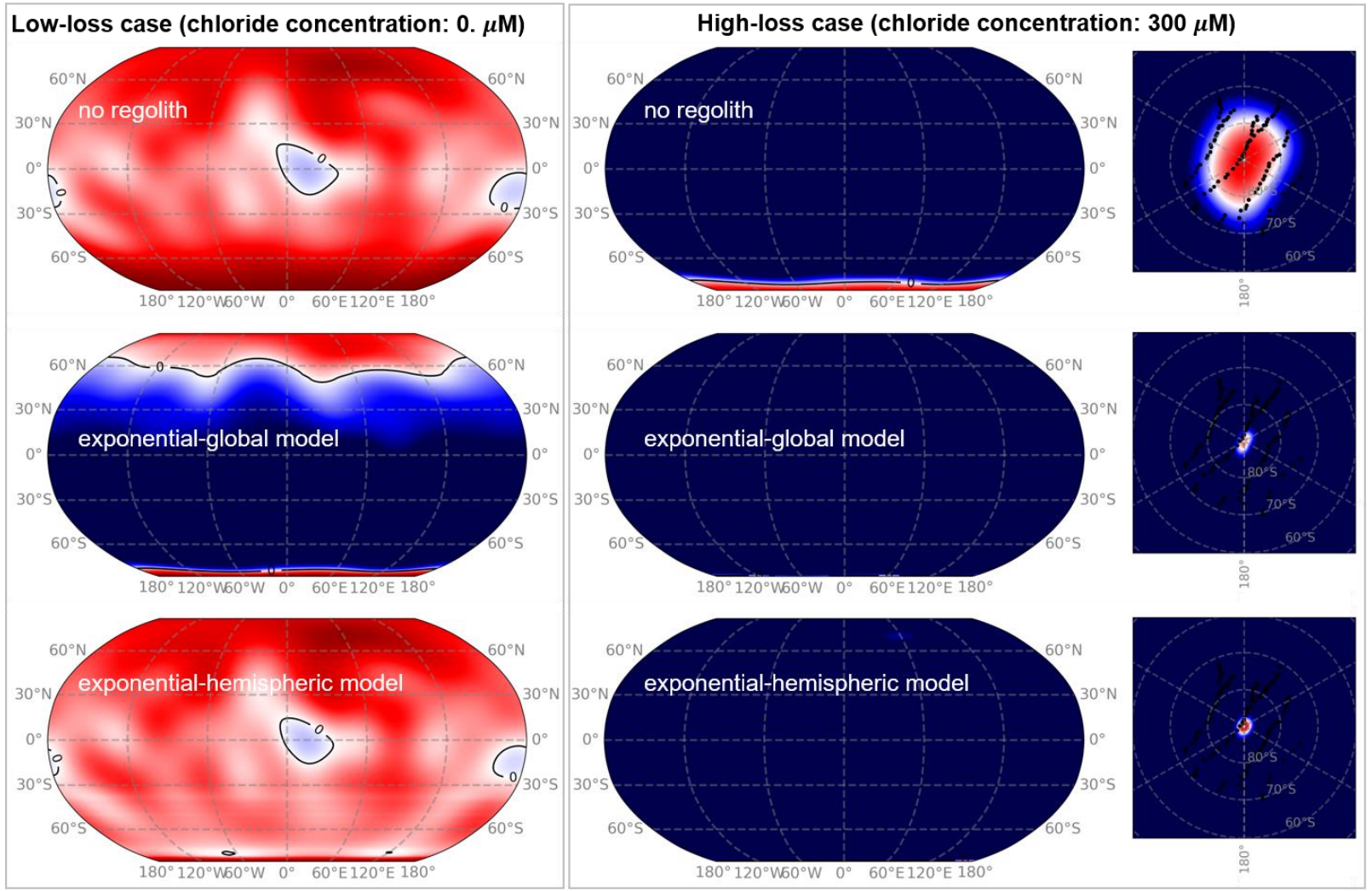
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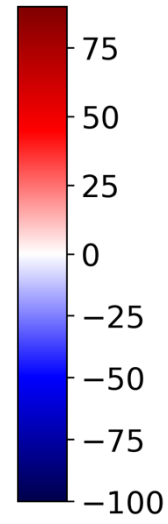
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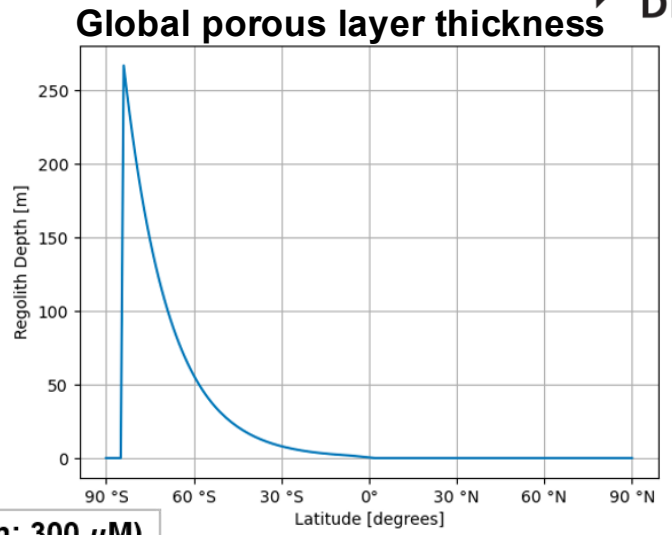




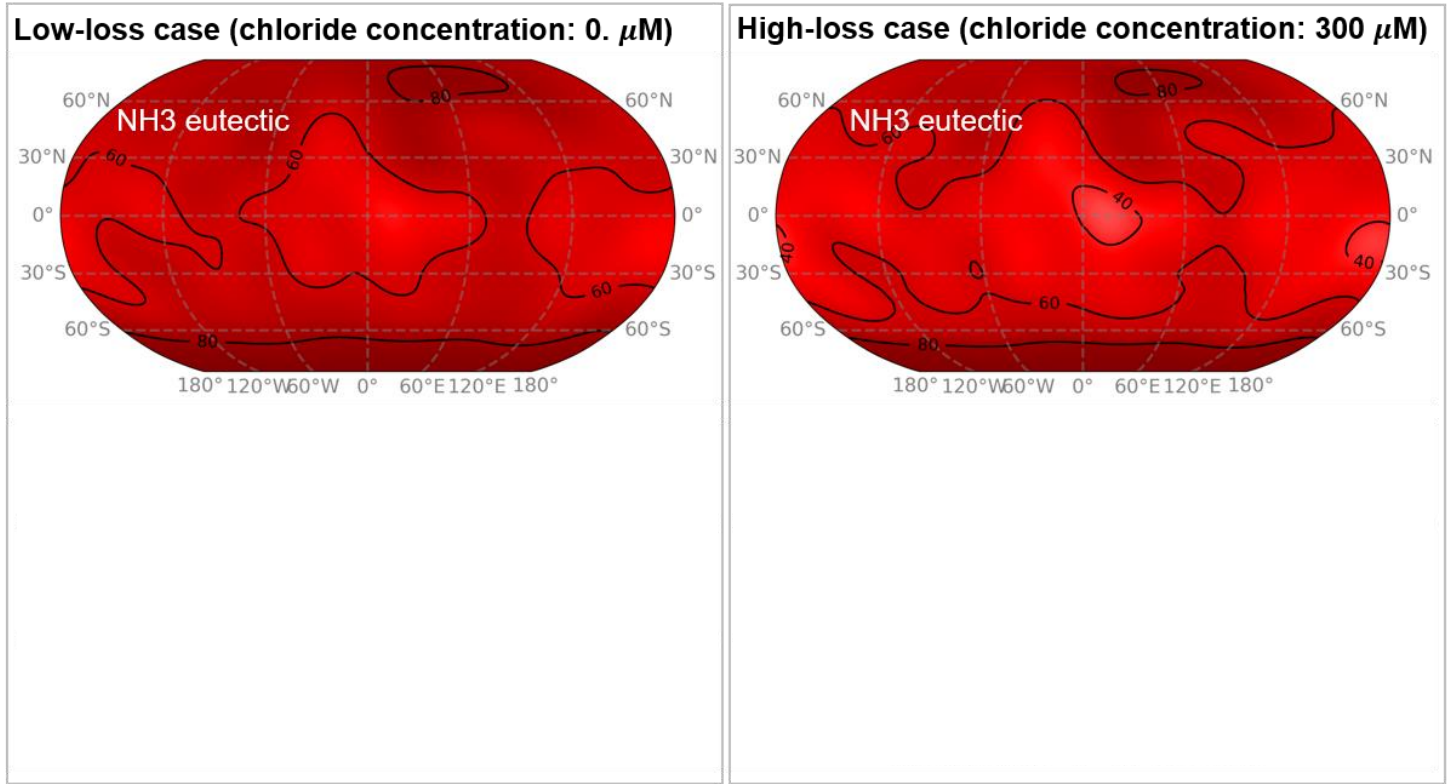
*assuming avg transmit power of 1.25 W and 100 km altitude orbit

Detectability of brines

- Brines may form within the ice shell when the ice shell temperature reaches the eutectic temperature
- They could provide a radar-detectable source
  - Eutectic interface of ammonia (NH_3): 175.45 K isotherm

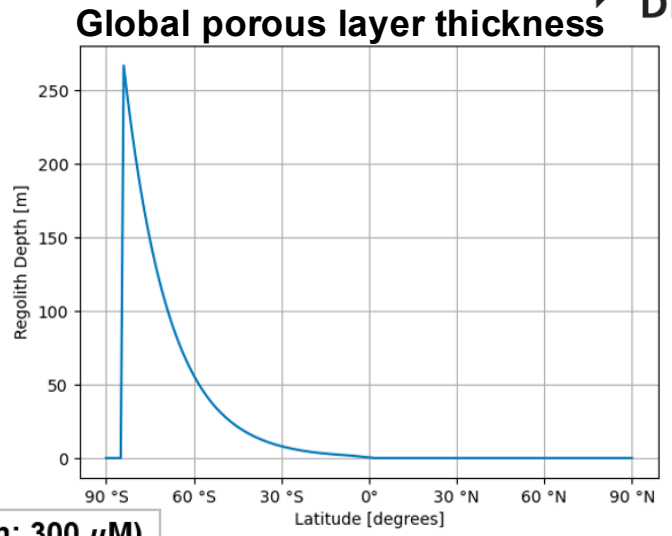


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  - Eutectic interface of ammonium chloride (NH_4Cl): 257.79 K isotherm

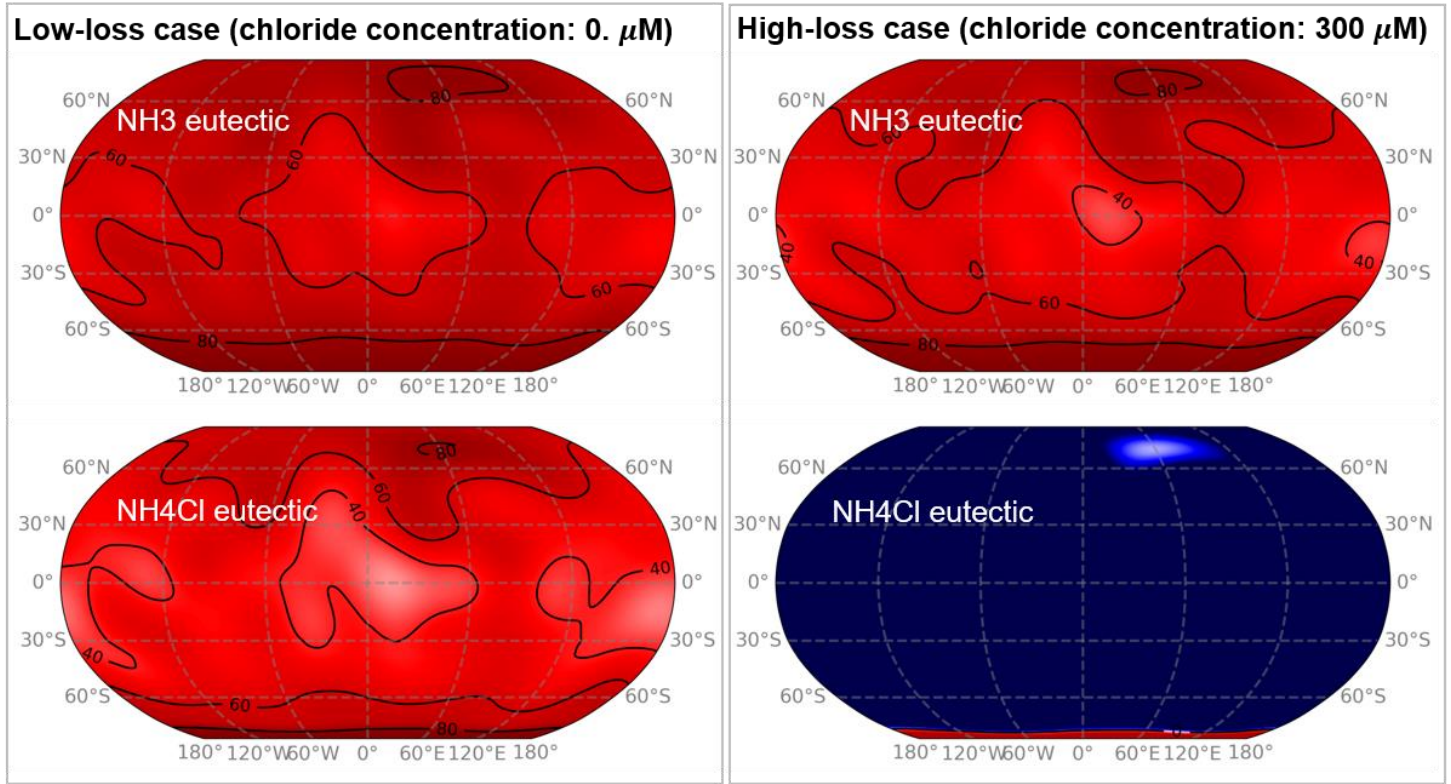


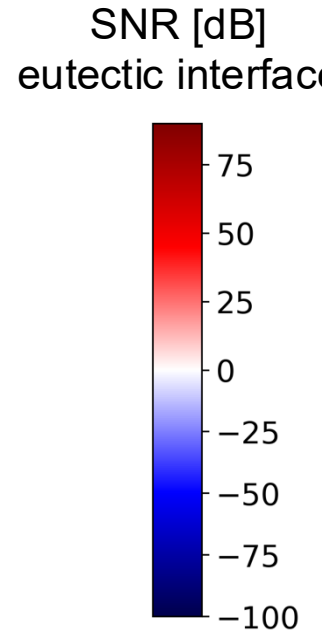
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See posters:

X4.126
Benedikter et al.

X4.129
Parihar et al.





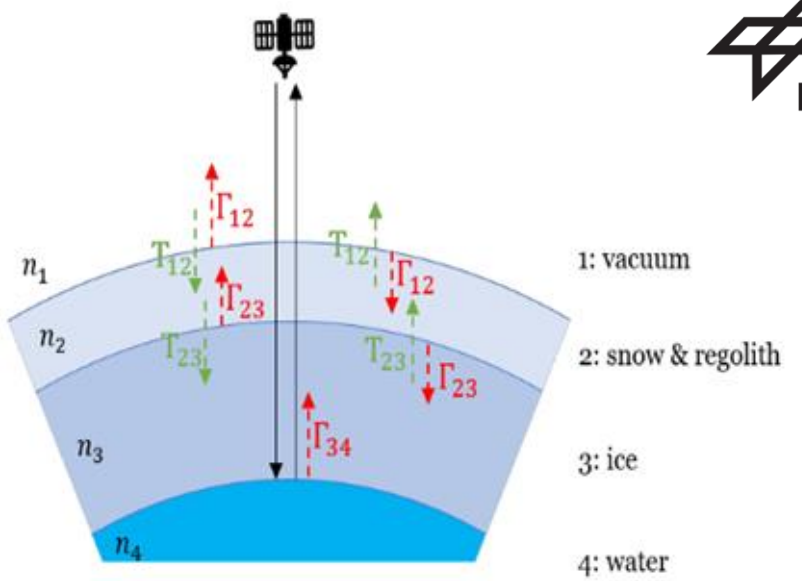
Conclusions

- The porous layer thickness and its distribution have a first order effect on the 2-way radar attenuation in Enceladus' ice shell.

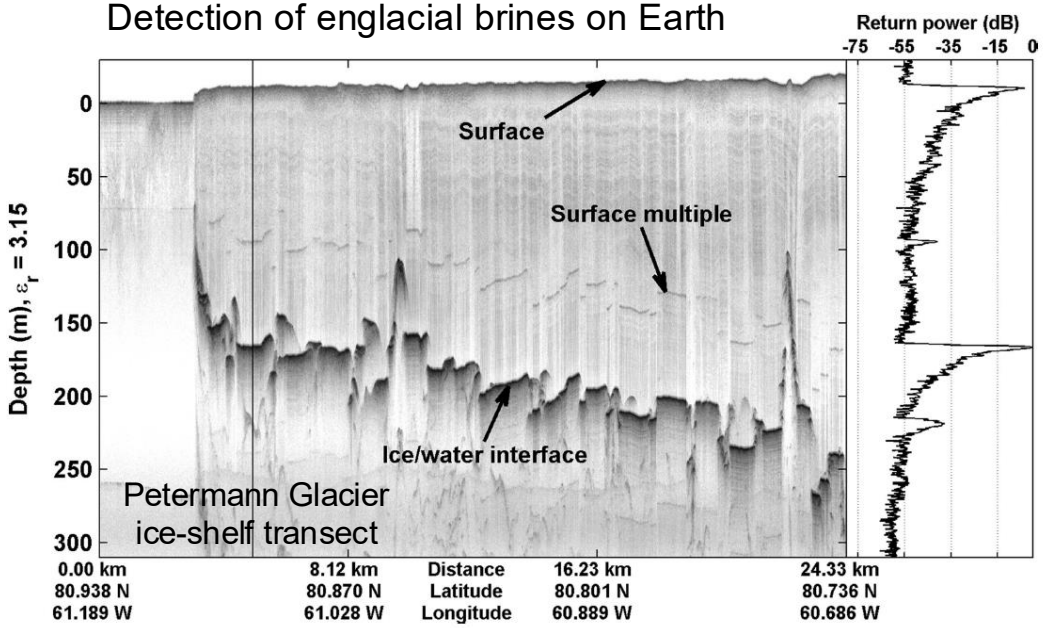
- Regions covered by a thick porous layer may prevent the radar signal from reaching the ice-ocean interface.

- However, the high temperatures may lead to the formation of shallow brines that would be detected by the radar measurements.

- Such brine reservoirs are considered niches for life, and their potential to be directly accessible by future missions is much greater compared to that of the oceans.



Detection of englacial brines on Earth

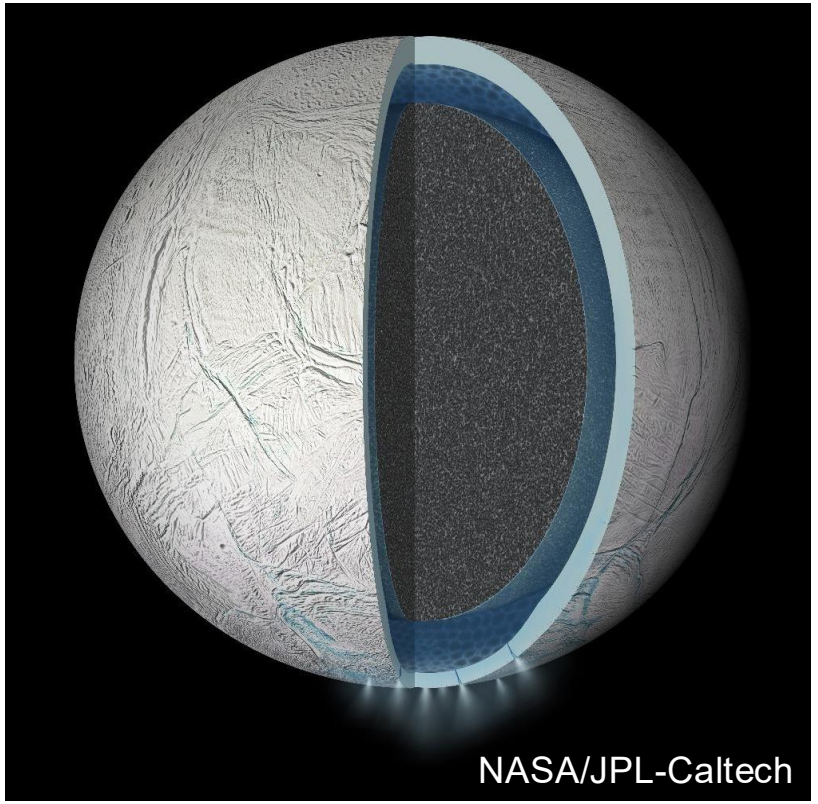


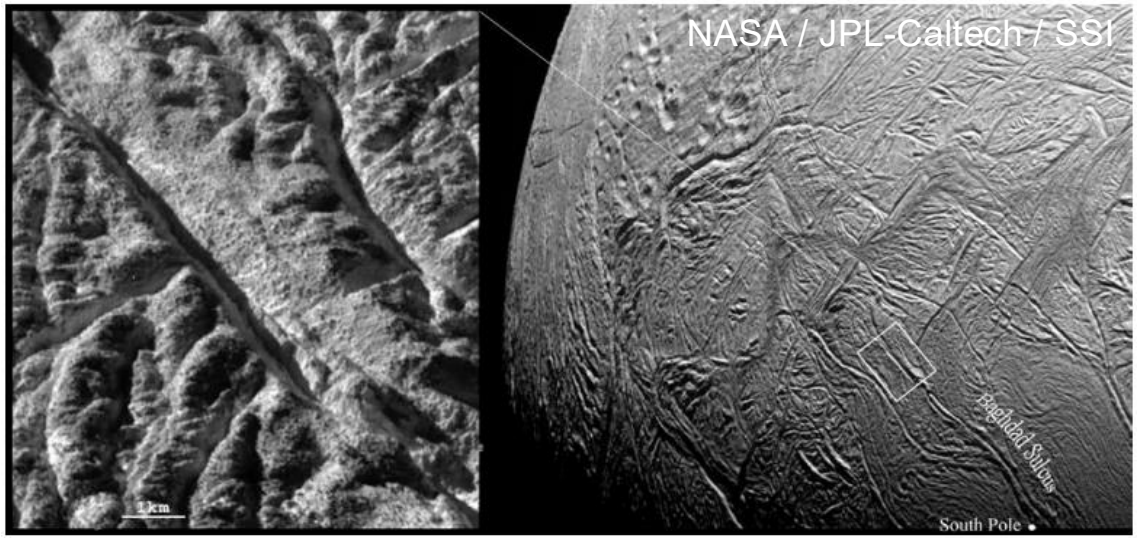
Lewis et al. 2015

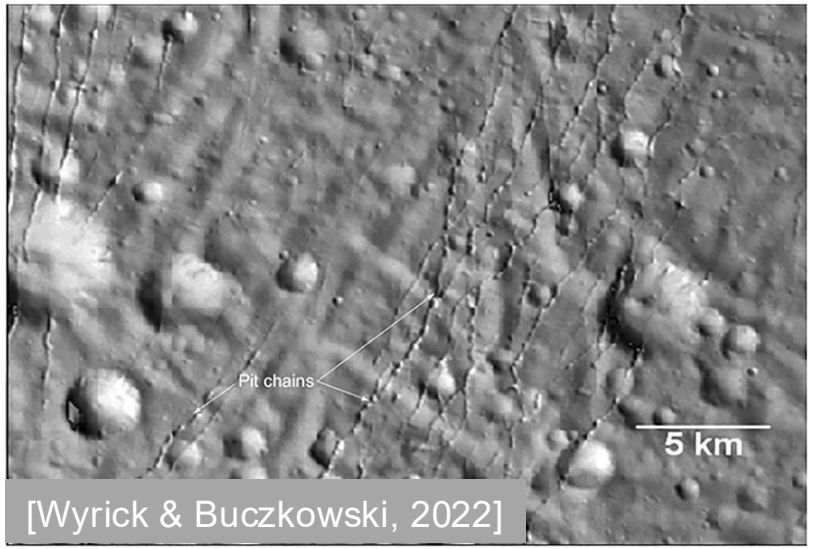
Additional Material

Ice Shell Stratigraphy

- The thin ice shell at the poles attributed to tidal heating
- Porous layer caused by deposition of material from Enceladus' plume
- Porous layer thickness from pit chains analysis
- Porous layer regional variations (boulders south pole)

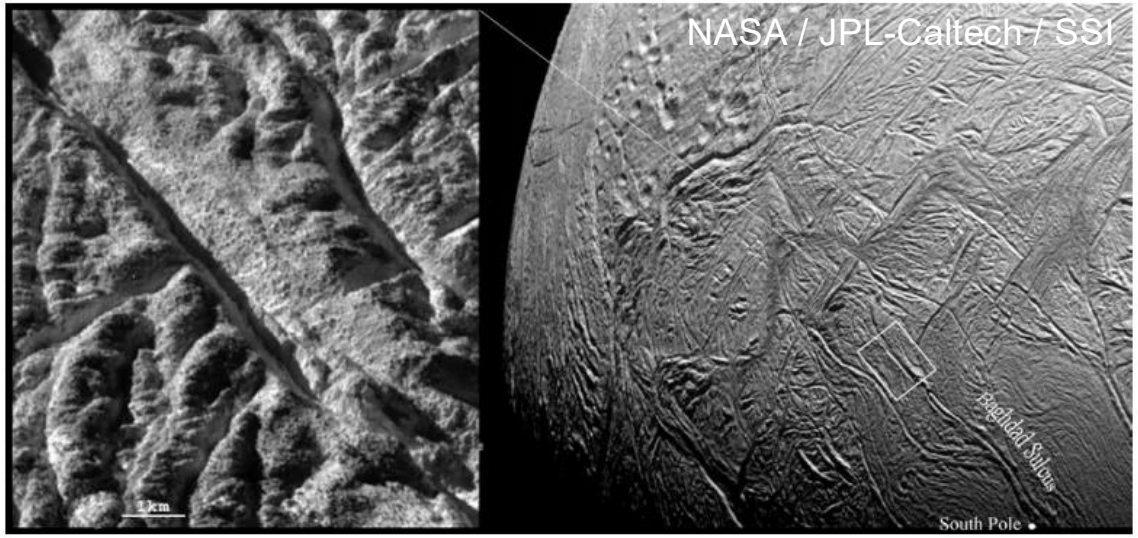


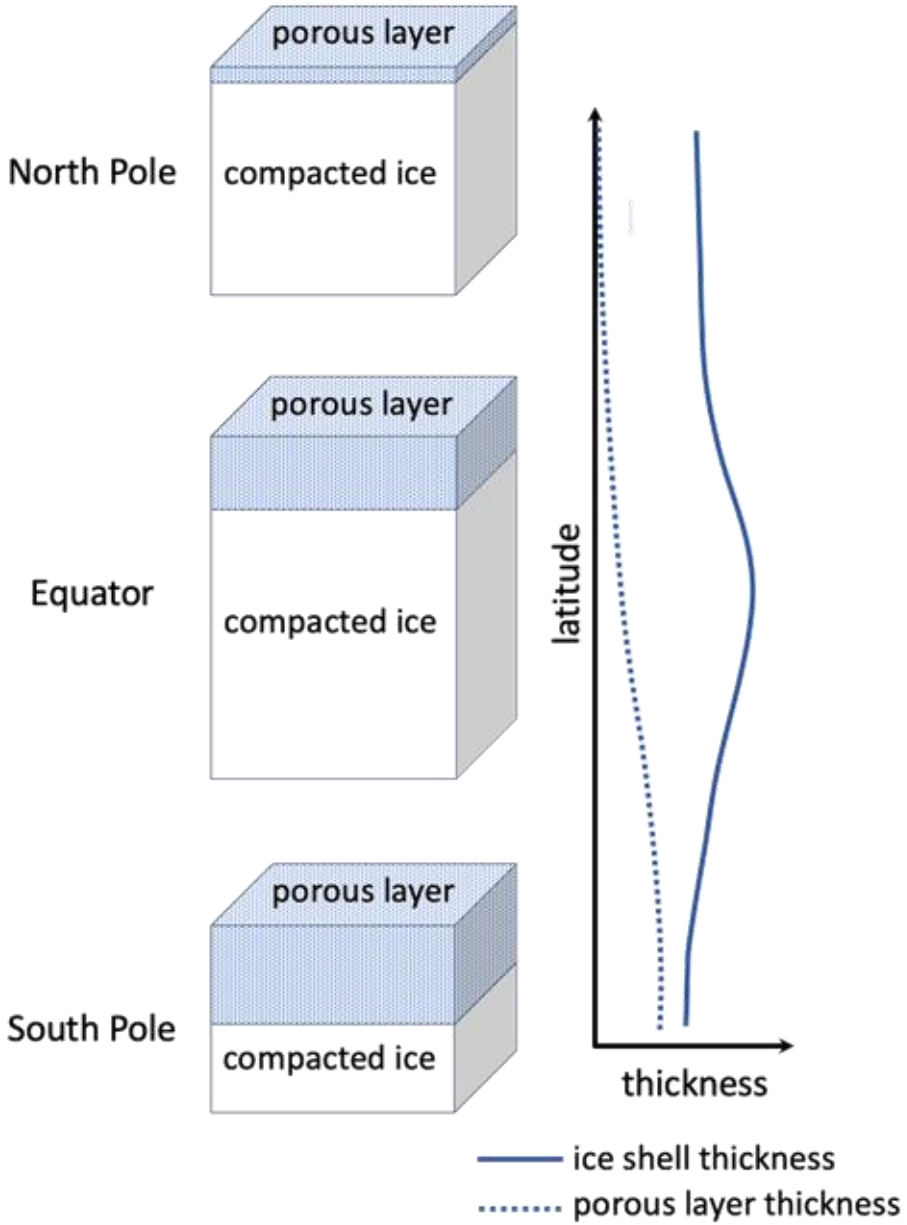




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Model parameters

- Thermal state calculated via the thermal diffusion equation:

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(k r^2 \frac{\partial T}{\partial r}\right) = 0$$

$$k_{ice} = \frac{651}{T}$$

[Petrenko & Whitworth, 1999]

$$k_{\text{porous layer}} = 0.001 - 0.1\,\mathrm{W\,m^{-1}\,K^{-1}}$$

[Seiferlin et al., 1996; Ferrari et al., 2021]

- Electrical conductivity depends on the thermal state:

$$\sigma = \sum_{i=0}^{1} \sigma_i^0 C_i \exp\left[-\frac{E_i}{k_B}\left(\frac{1}{T} - \frac{1}{T_r}\right)\right]$$

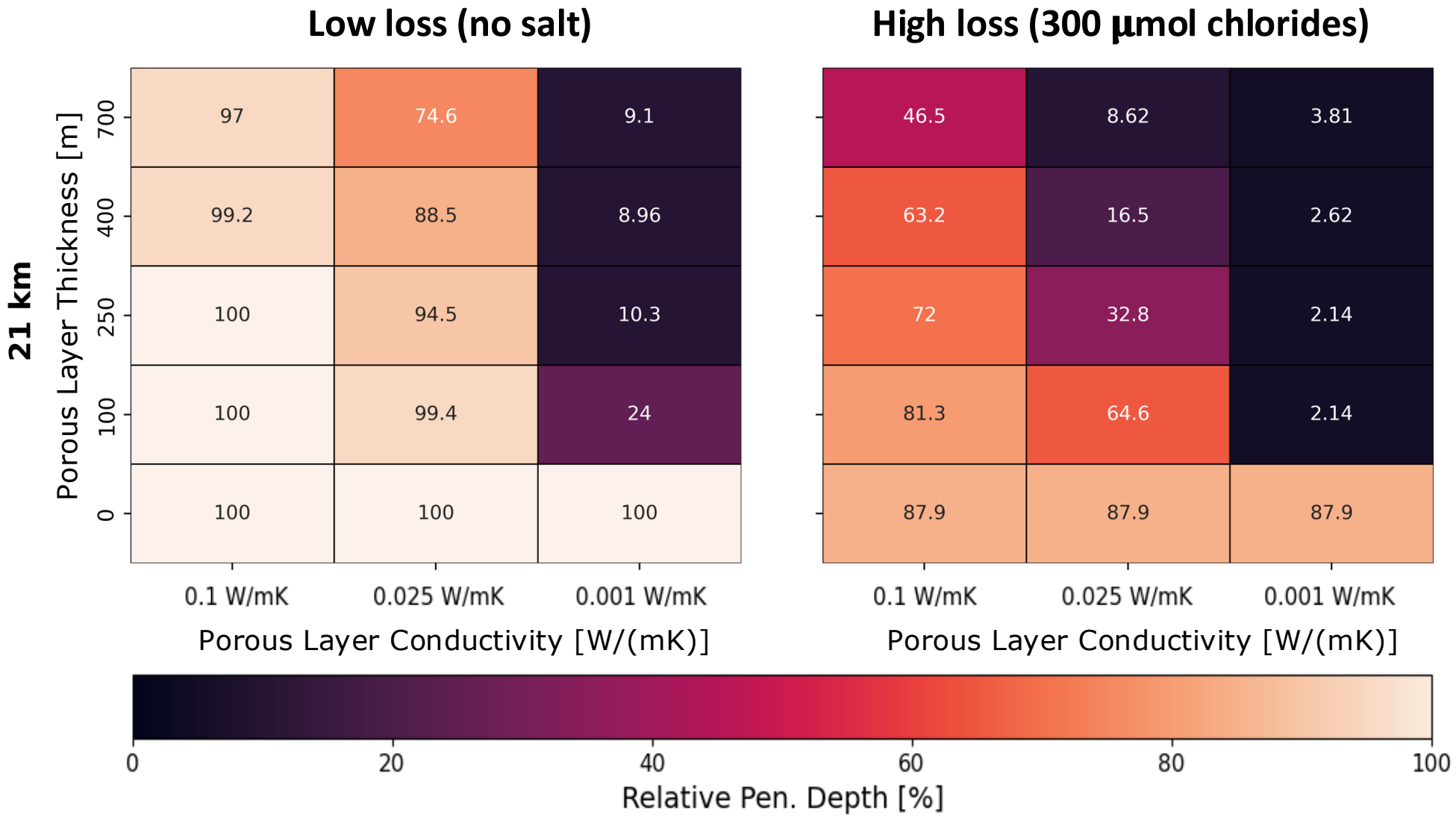
[Kalousová et al., 2014; Souček et al., 2023]

- 2-way radar attenuation using the electrical conductivity:

$$A_1 \approx 0.914\sigma \qquad A_2(d) = \sum_{i=0}^{N} 2A_1(d_i)\Delta d$$

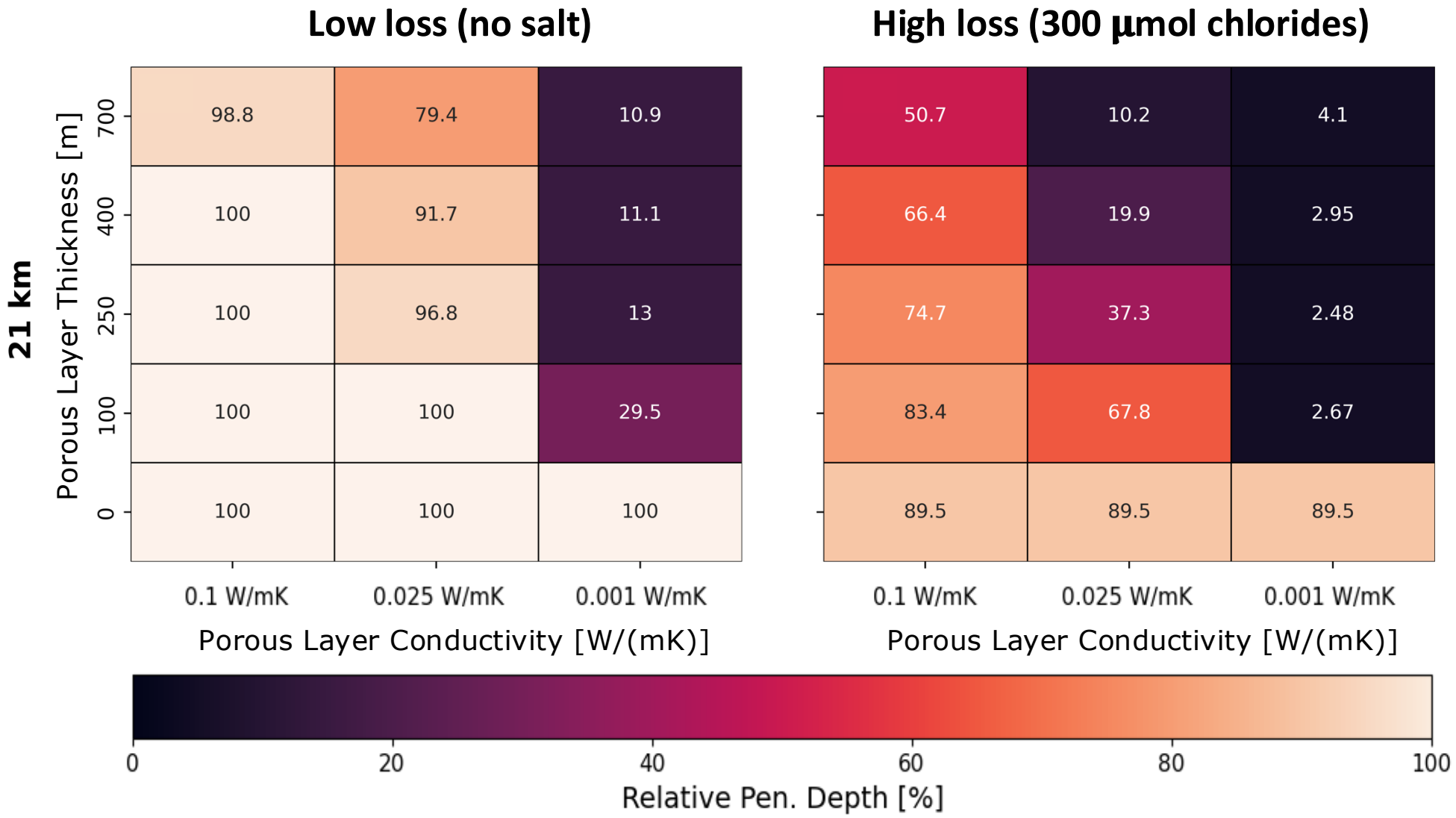
2-way attenuation in a 21 km ice shell

Attenuation limit: 100 dB



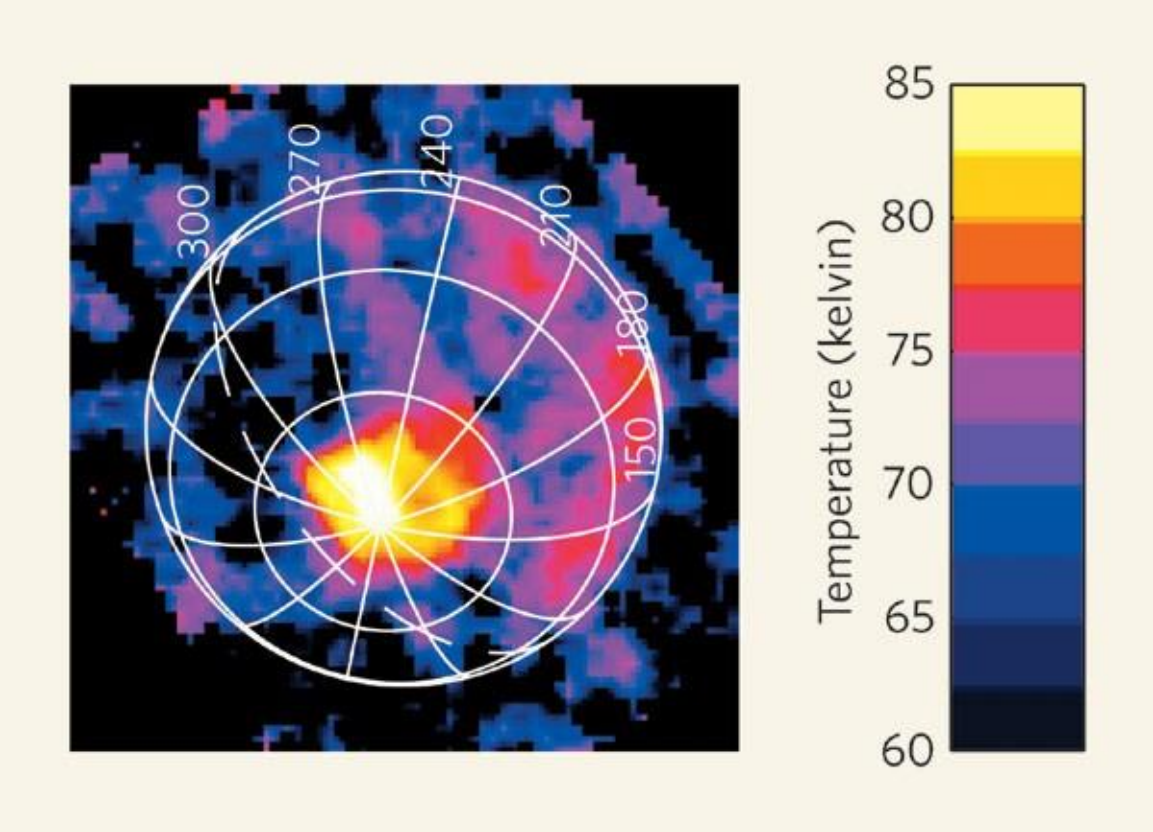
2-way attenuation in a 21 km ice shell

Attenuation limit: 130 dB

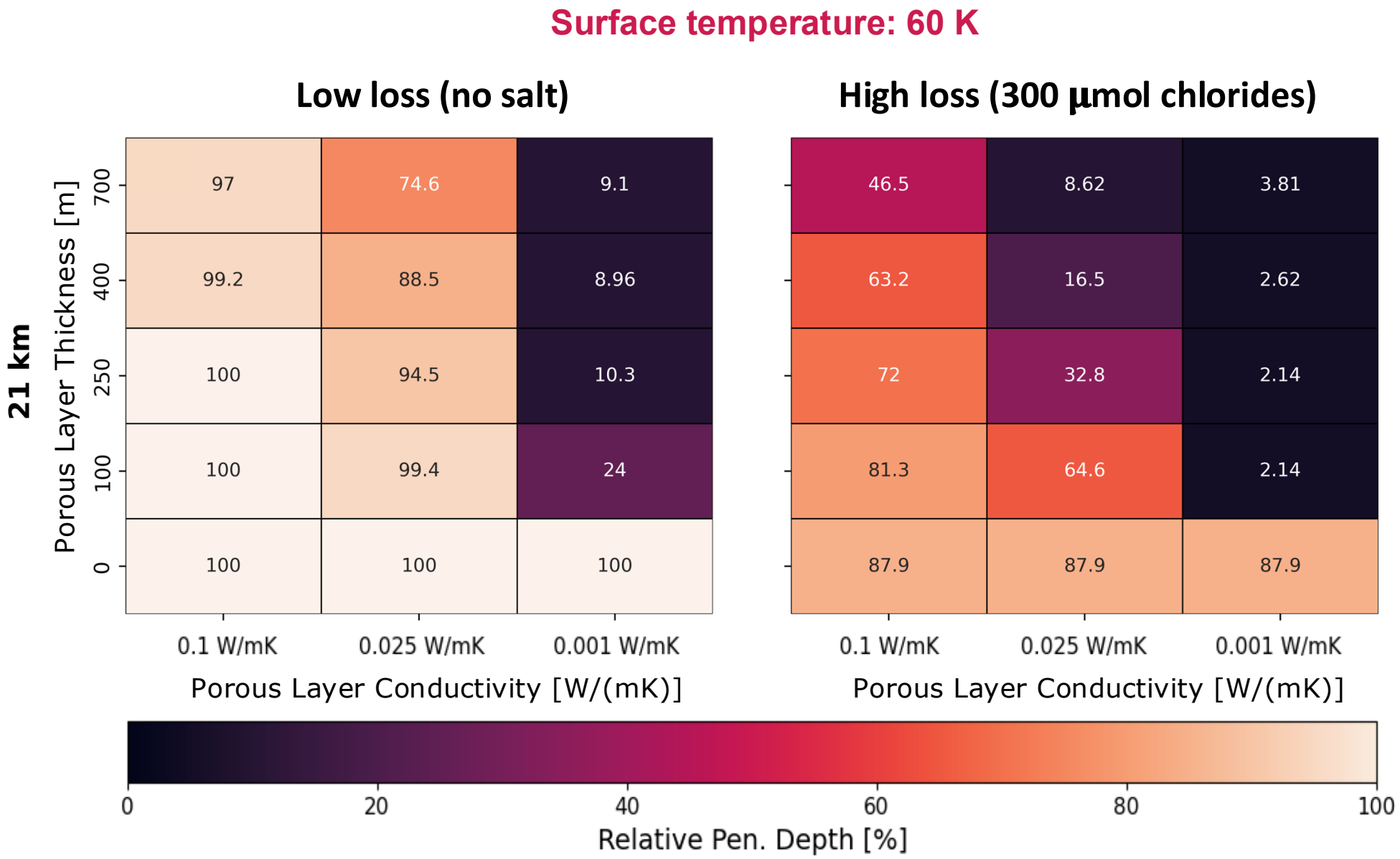


Effects of surface temperature on 2-way attenuation

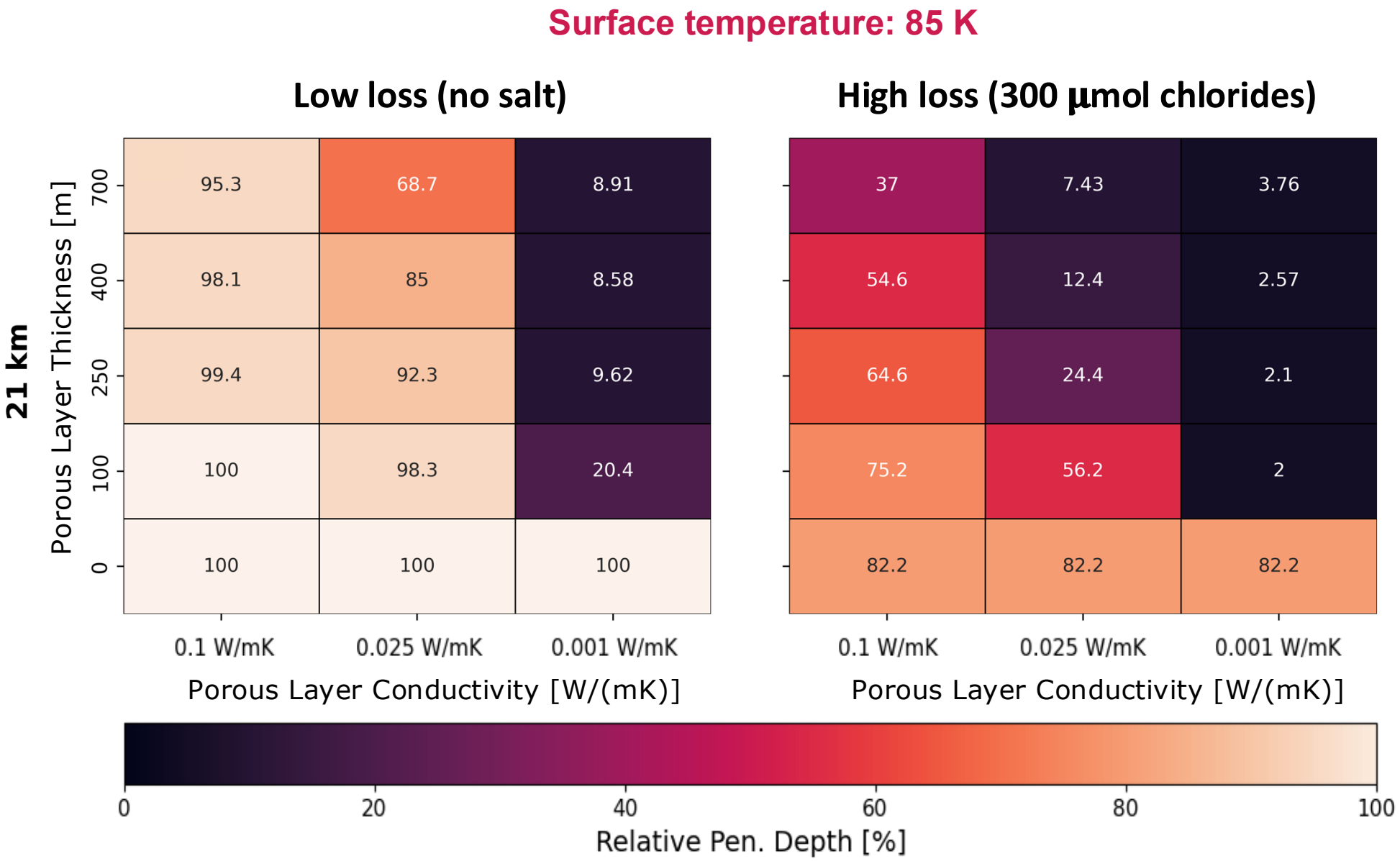
- The surface temperature on Enceladus varies
- The plot show the intensity of heat radiation (10- to 16-micron wavelength range) translated into temperature
- Peak south polar temperature ~85 K averaged over the 30-kilometer spatial resolution of the data
- The distribution of heat radiation suggests that most or all of the south polar heat comes from the tiger stripes



2-way attenuation in a 21 km ice shell

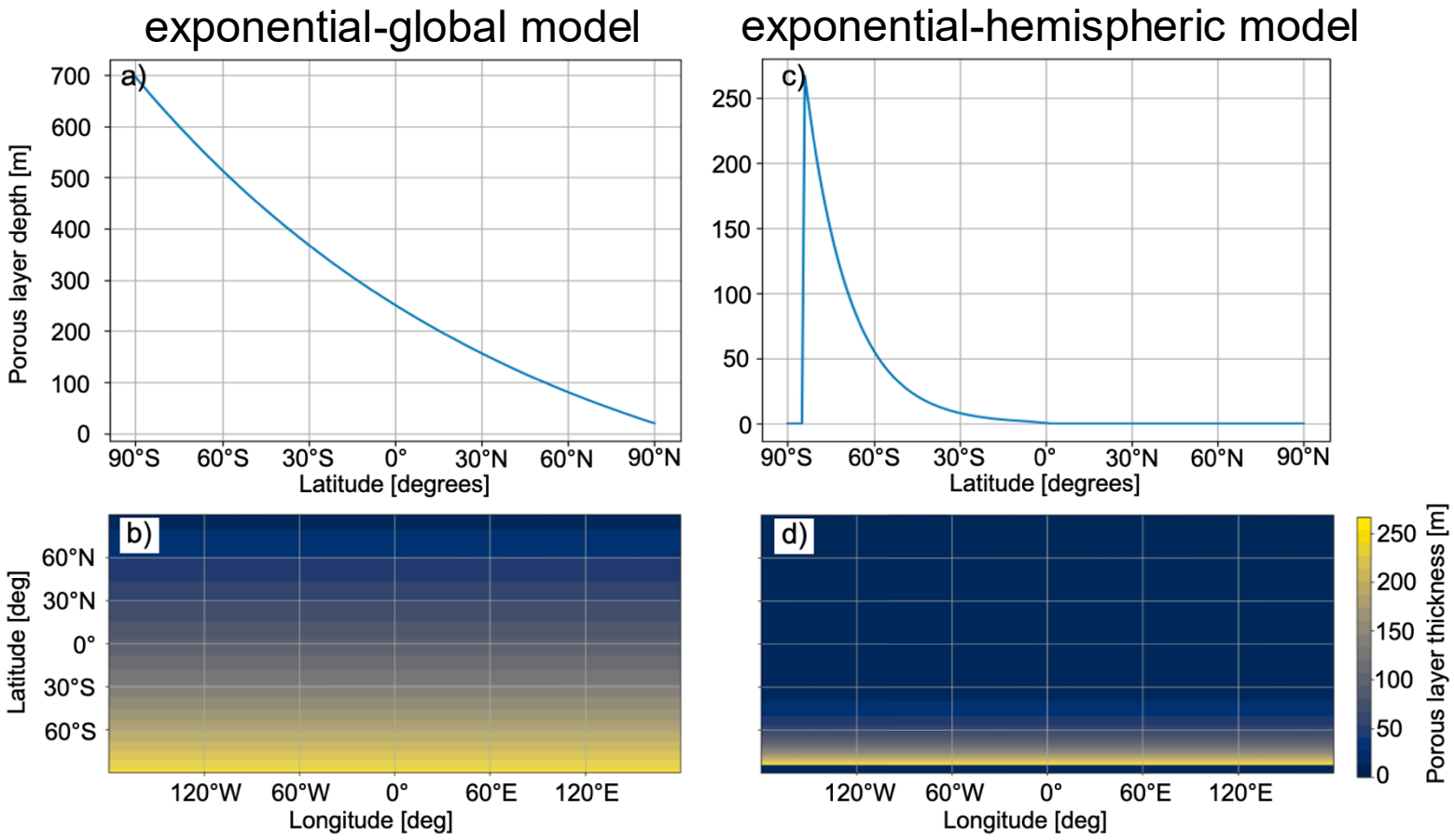


2-way attenuation in a 21 km ice shell



Enceladus: Ice Shell Thickness Variations

- The ice shell thickness: obtained from shape*, gravity and libration data
- The values depend on the chosen shape model and the assumed density difference between ice shell and ocean
- Ice shell thickness:
  - Average value: 21-23 km
  - South pole:
    values as low as 5 km
  - Equatorial regions:
    values as high as 35 km
- Porous layer:
  - Thickness up to 700 m
  - Distribution poorly constrained



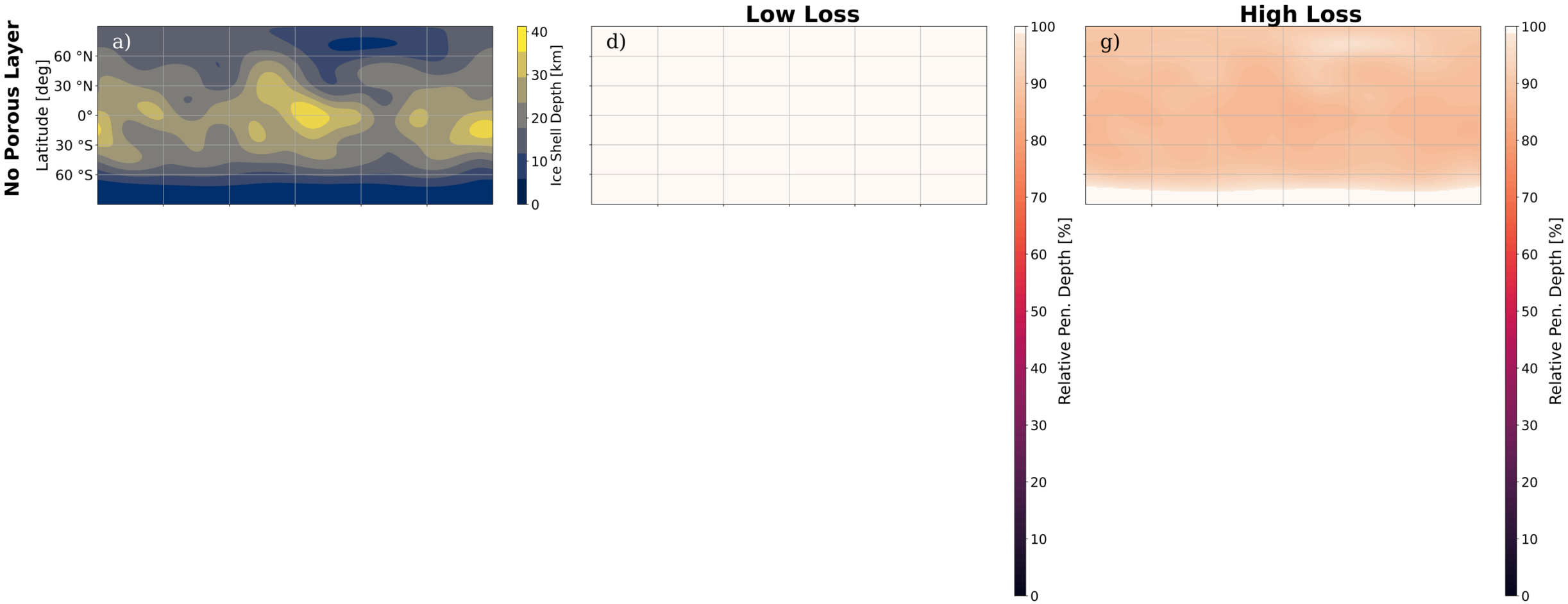
[Byrne et al., submitted to JGR:Planets]

Effects of Porous Layer Distribution on 2-way Attenuation

- Ice shell thickness estimates obtained from shape, gravity and libration data
- Porous layer thickness estimates from the analysis of pit chains

[Hemingway & Mittal, 2019]

[Martin et al., 2023]



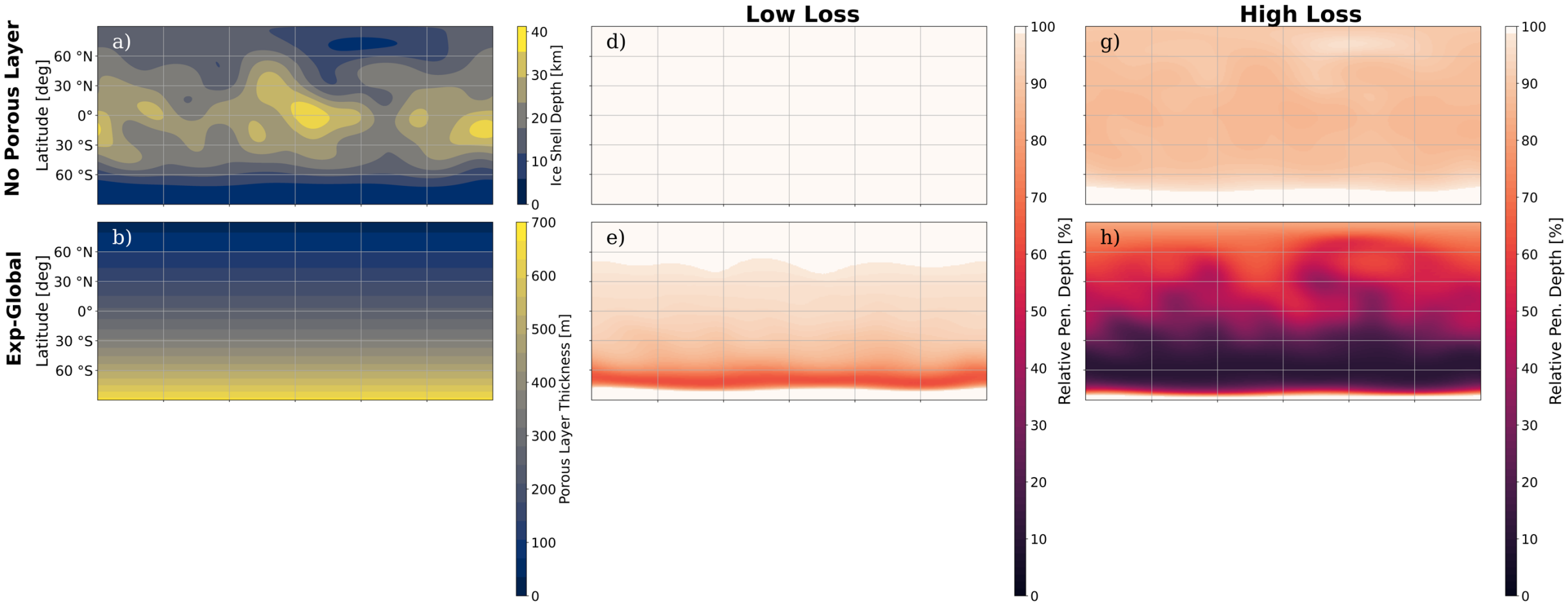
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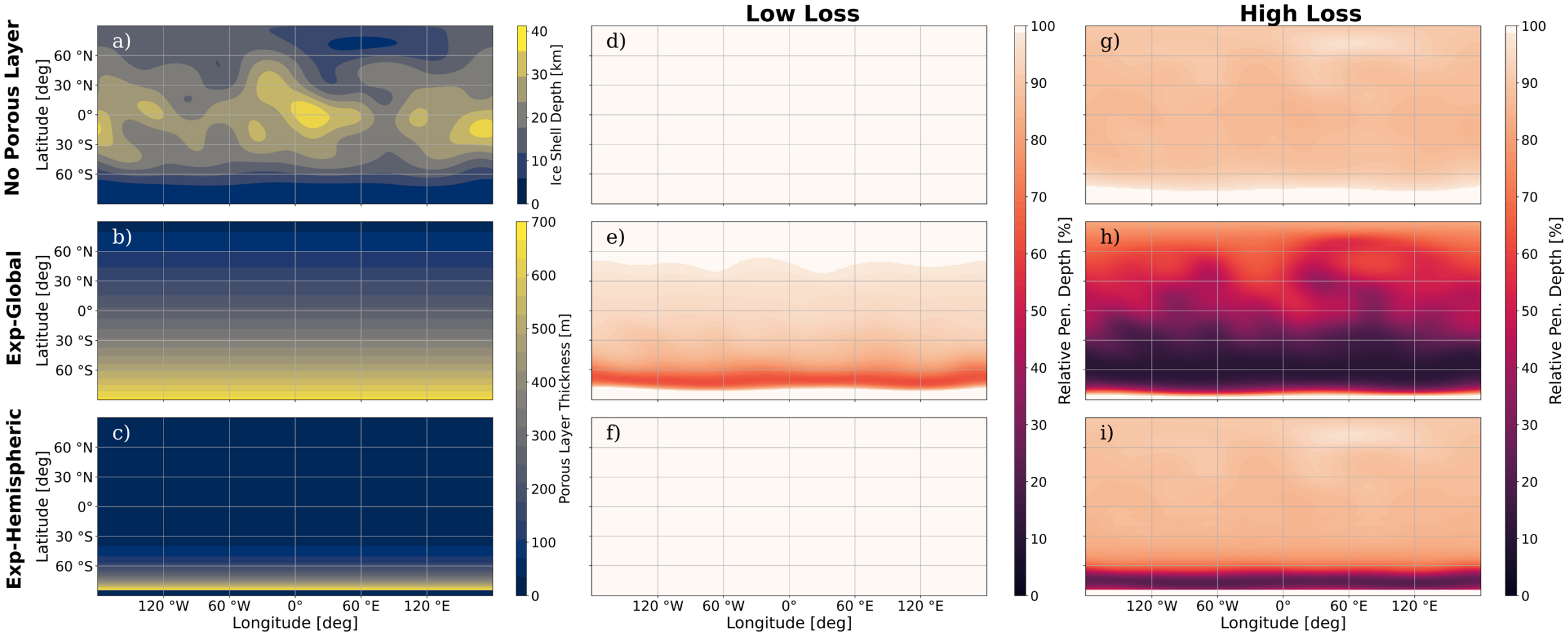
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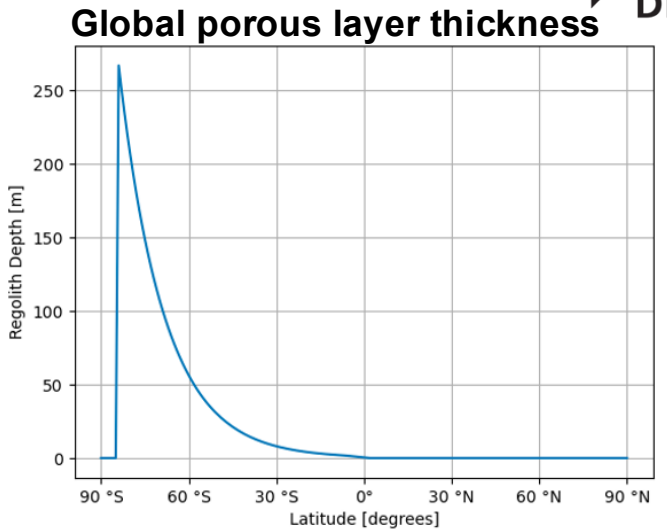
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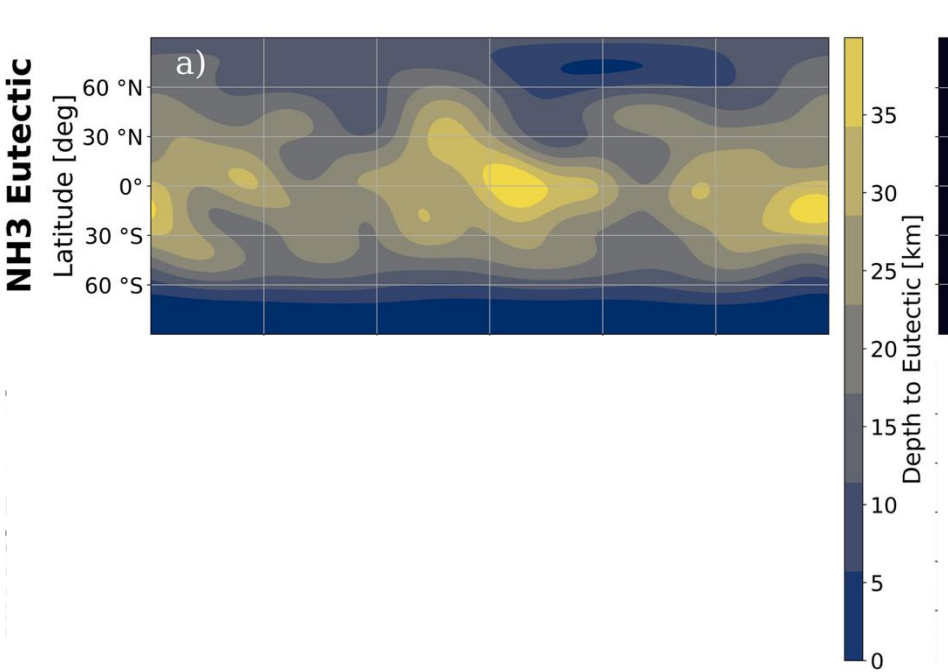


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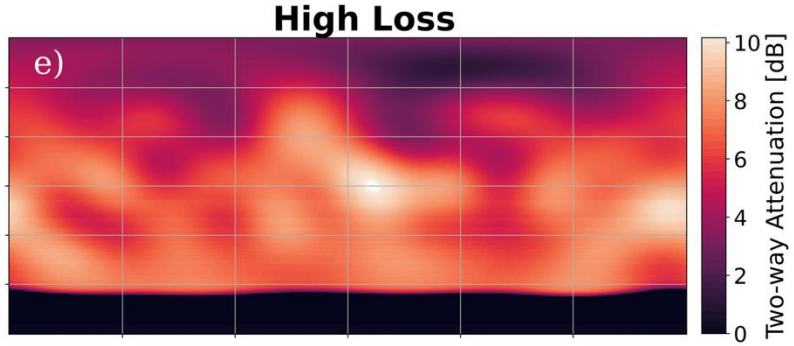
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[Byrne et al., submitted to JGR:Planets]

Detectability of brines

- Brines may form within the ice shell when the ice shell temperature reaches the eutectic temperature
- They could provide a radar-detectable source
  - Eutectic interface of ammonia (NH_3): 175.45 K isotherm
  - Eutectic interface of ammonium chloride (NH_4Cl): 257.79 K isotherm

